

exactly 80 mm: _____

FIRST SOLVE

New here? Do the walkthrough once at cubepath.com/5x5-vercel.app/

This cube is the memory jog:

NOTATION

R U L D B = turn that face clockwise, looking at from outside the cube

r = counter-clockwise, z = half turn, Lowercase r = that face plus the layer behind it, turning together.

y = spin the whole cube left, white staying down. A whole-cube turn solves nothing.

WORDS

algorithm = a fixed sequence of turns - trigger = a short algorithm your hands run as one unit

1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm - work it out. Can't see it? Build the four white edges on the YELLOW face first, then solve its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top, turn U until it sits above the yellow slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT
RU'R'



white sticker FRONT
LU'L



white sticker UP run the first one once to knock it out, then look again
RU'R'

3 MIDDLE EDGES

Top edge with NO yellow: turn U until its front color matches the center it belongs to. The arrow shows which slot it goes to.

goes **RIGHT**
U **RU'R'** y **LU'L**



goes **LEFT**
U **L'** **LU'L** y' **RU'R'**

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

■ RU'R sexy move
 ■ RU'R Sune
 ■ R'FRP' sledgehammer
 gap = change grip

FIRST SOLVE

New here? Do the walkthrough once at cubepath-six.vercel.app/ learn. This is the memory jog.

NOTATION

R = turn that face clockwise, looking at it from outside the cube
c = counter-clockwise, **z** = half turn. Lowercase **r** = that face plus the layer behind it, turning together.
y = spin the whole cube left, white staying down. A whole-cube move does nothing.

WORDS

algorithm = a fixed sequence of turns - trigger = a short algorithm your hands run as one unit

1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's color must match the center it touches. No algorithm - work it out. Can't see it? Build the four white edges on the YELLOW face first, then solve its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top, turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT
R'UR'



white sticker FRONT
L'UL



white sticker UP run the first one once to knock it out, then look again
R'UR'

3 MIDDLE EDGES

Top edge with NO yellow: turn U until its front color matches the center under it. The arrow shows which slot it goes to.

goes **RIGHT**
U R'UR' y L'UL



goes **LEFT**
U L'UL y' R'UR'

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

■ **RUR** sexy move
 ■ **RUR** Sune
 ■ **R'FR'** sledgehammer
 gap = change grip

TWO-LOOK

OLL CROSS - yellow edges



Line bar left-to-right
fRUR'U'f



Hook hook pointing front-right
fRUR'U'f

OLL CORNERS - yellow face



Sune $\frac{1}{2}$ yellow, rest clockwise
RUR'U RU2R'



Anti-Sune $\frac{1}{2}$ yellow, rest counter-clockwise
RU2R' U'RUR'



P1 $\frac{1}{2}$ yellow; headlights left
fRUR'U'f fRUR'U'f

OLL CORNERS - continued



Headlights $\frac{3}{4}$ 2 yellow; headlights at the back
R2DR'U2 RD'R'U2R'



2x Headlights $\frac{3}{4}$ 0 yellow; headlights both sides
RUR'U RU'R' URU2R'



Chameleon $\frac{3}{4}$ 2 yellow next to each other
rRUR'U' r' FRF'



Bowie $\frac{3}{4}$ 2 yellow diagonal
fR'UR'U' r' FR

FAILBACK

The badge on each row is your failback: that many Sunes, with a U turn around, also finishes the case. Machine-checked, there is the worst.
Top not all yellow and nothing matches? Run Sune and read again.

No yellow edge at all? Run the first one, then read again.

■ RUR'U sexy move
 ■ RUR'U Sune
 ■ R'FRF' sledgehammer
 gap = change grip

TWO-LOOK

OLL CROSS - yellow edges



Line bar left-to-right
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OLL CORNERS - yellow face



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OLL CORNERS - continued



Headlights $\frac{1}{2}$ yellow; headlights at the back
R2DR'U2 RD'R'U2R'



2x Headlights $\frac{1}{2}$ yellow; headlights both sides
RUR'U RUR'R URU2R'



Chameleon $\frac{1}{2}$ yellow next to each other
r'RUR'r' FRF



Bowtie $\frac{1}{2}$ yellow diagonal
f'RUR'U'r' FR

FAILBACK

The badge on each row is your fallback: that many Sunes, with a U turn around, also finishes the case. Machine-checked; there is the worst.
Top not all yellow and nothing matches? Run Sune and read again.

No yellow edge at all? Run the first one, then read again.

■ **RUR'U** sexy move ■ **RUR'U** Sune ■ **R'FR** sledhammer ■ **gsp** = change grip

ONE-LOOK PLL

ADJACENT CORNER SWAP

exactly one face shows headlight



***T1, headlights; pairs B-left F-left**
RUR'U' R'F R2UR'U' RUR'F'

Ja L solid; pairs B-right F-left B-front
x R2FRF' RU2' rUr U2

Jb L solid; pairs B-left F-back
RUR'F' RUR'U' R'F R2UR'U'

Aa L headlights; pairs F-right B-front
x L2D2 L'UL' D2 L'UL'

Ab L headlights; pairs B-right B-back
x' L2D2 LUL' D2 LU'L

F L solid
R'U'F' RUR'U' R'F R2UR'U' RUR'UR

ADJACENT CORNER SWAP

continued



Ra L1, headlights; pairs F-left
RU'R'U' RUR DR'UR'D' R'U2R'



Rb L1, headlights; pairs B-left
R2F RURU' FR U2RU2R



Ga L1, headlights; pairs F-right
R2UR'UR'U' RU'R2 U'D R'UR'D'



Gb L1, headlights; pairs B-back
RUR' UD' R2UR'UR'U' RU'R2D



Gc L1, headlights; pairs B-right
R2UR'RU' RU'R2 UD' RU'R'D



Gd L1, headlights; pairs B-front
RUR' UD' R2UR'UR'U' UR'U2R2

Learn in the printed order, three a week, about six weeks. Ja and Jb on the same day – one is the mirror of the other.

ONE-FACE LOOK PLI ADJACENT CORNER SWAP	ADJACENT CORNER SWAP continued
 T L solid; pairs B-left F-left RUR'U' R'F R2UR'U' RUR'F'	 Ra L' headlight; pairs F-left RU'R'U' RU'R DR'U'R'D' R'U2R'
 Ja L solid; pairs B-right F-left R-front x R2FRF' RU2 r'Ur U2	 Rb L' headlight; pairs B-left R2F RURU'R' F'R U2R'U2R
 Jb L solid; pairs B-left F-right R-back RUR'F' RURU'R' R'F R2UR'U'	 Ga L' headlight; pairs F-right R2UR'UR'U' RU'R2 U'D R'UR'D'
 Aa L' headlight; pairs F-right B-front x L2D2 L'U'L D2 L'U'L	 Gb L' headlight; pairs B-back RUR'U'D R2UR'URU' RU'R2D
 Ab L' headlight; pairs B-right R-back x' L2D2 LUL' D2 LU'L	 Gc L' headlight; pairs B-right R2UR'RU' RU'R2 UD' RU'R'D'
 F L solid RU'F' RUR'U' R'F R2UR'U' RU'RUR	 Gd L' headlight; pairs R-front RUR' U'D R2UR'RU' RU'R2U'D'

ANNEX – BIG CUBES AND THE MAP

BIG CUBES 4x4 and 5x5
Before I build the 6 centers → pair the 12 edges → solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2nd Layer ONLY – never widen the cube.

last 2 edges

4x4 PLL parity

$$\begin{array}{ccccccc} R'w & U' & R' & F & R' & F' & Rw \\ 2R2 & U2 & 2R2 & Uw2 & 2R2 & U2 \end{array}$$

slicing cut, run it, slice back – both cubes
2 edge pairs swapped - half the time

4x4OLL parity = 1 edge pair flipped- half the time

$$R'w \ U2 \ x \ R'w \ U2 \ Lw \ U2 \ R'w \ U2 \ Lw \ U2 \ R'w \ U2 \ Rw \ U2 \ Lw \ U2 \ Rw'$$

R's edge twisted → edge not twisted → the same move differs, so swap PLL parity.

R's edge twisted → edge not twisted → the same move differs, so swap PLL parity.

$$R'w \ U2 \ Lw \ U2 \ R'w \ U2 \ Lw \ U2 \ R'w \ U2 \ Lw \ U2 \ R'w \ U2 \ Rw' \ U2 \ Rw' \ U2 \ Rw'$$

Fix parity during the last layer, before OLL and PLL – both parity algorithms move corners.

NOTATION

R U L D B = turn that face clockwise, from outside the cube.'s
outside clockwise = 2 half turns.
M = middle slice, turning the way i turns. x y z = the whole cube,
turning the way R or L or B over twice i + is that plus the flip
behind it, turning together.

WORDS

algorithm = a fixed sequence of turns - trigger = a short algorithm your hands runs as one unit - solve move → the R solve move the R → trigger.
the trigger = the P or F or trigger, parities i + is a 3x3 cannot have edge pair → two edges glued into one, on a 4x4 or 5x5

RURU's sexy move **RURU**SuSu **FRFP** sledhammer gap = change grip

DDD A-A

Twist a yellow corner four turns at a time, turning ONLY the top
corner clockwise.

yellow faces RIGHT **D'R'D** x2

yellow faces FRONT **D'R'DR** x2

ANNEX – BIG CUBES AND THE MAP

BIG CUBES 4x4 and 5x5

Before we build the 6 centers → pair the 12 edges → solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2nd LAYER ONLY – never mind.

last 2 edges

4x4 PLL parity

4x4 OLL parity – 1 edge pair flipped - half the time

Rw U2 x Rw U2 x Rw U2 R'w U2 Lw U2 R'w U2 Rw U2 R'w U2 R'w
5x5 edge parity – edge colored pairs – 1 move more difficult, so 5 PLL parity

5x5 edge parity – 1 edge swapped – 1 move more difficult, so 5 PLL parity

Few parity during the 1st layer, before OLL and PLL – both parity algorithms move corners.

NOTATION

R U F D B + turn that face clockwise, from outside the cube. '⁻'
= counter-clockwise, 2 = half turns.

M = middle slice, turning the way I turns. x y z = the whole cube,
turning the way I turns. F = flip. Overcase r f = that plus the flip
layer behind it, turning together.

WORDS

algorithm = a fixed sequence of turns - trigger = a short algorithm your hands run as one unit - setup move - the R U or U' trigger
trigger name = the P-F-R trigger; parity = a case a 3x3 cannot have; edge pair = two edges glued into one, on a 4x4 or 5x5

RURU' RURU' RURU' RURU' FRFP sledhammer gap+ change grip

Print at 100% / Actual size — do NOT fit to page. Two-sided, flip on the LONG edge. The number exactly 80 mm: _____

Print at 100% / Actual size — do NOT fit to page. Two-sided, flip on the LONG edge. The numeral on both faces is your check: cut ONE card first. This bar must measure exactly 80 mm: 

FIRST SOLVE 4 YELLOW CROSS -yellow edges only – ignore the corners  Dot on yellow edge at all FRUR'U'F' Hook two at a right angle: hold them BACK and LEFT FRUR'U'F' Line hold the bar left-to-right FRUR'U'F' One algorithm, up to three times: dot to hook to line to cross. 5 MATCH THE YELLOW EDGES First turn U so at least two top edges match the center under them. This algorithm is called Sune; you use it on every card after this one.  two matched, side by side hold them BACK and LEFT RUR'U' RU2R'  two matched, opposite run it once from any hold, then look again RUR'U' RU2R' LAST-LAYER ORDER ON THIS CARD: yellow cross, match edges, place corners, twist corners. Card 2 replaces this order permanently – it is the only thing on this card that gets retired. UNLOCK CARD 2 – five scrambles in a row with this card face down. ◻ MASTER: those five under 3:00. My target: _____. -NEXT: 2/3 TWO-LOOK  80 mm	FIRST SOLVE 4 YELLOW CROSS -yellow edges only – ignore the corners  Dot on yellow edge at all FRUR'U'F' Hook two at a right angle: hold them BACK and LEFT FRUR'U'F' Line hold the bar left-to-right FRUR'U'F' One algorithm, up to three times: dot to hook to line to cross. 5 MATCH THE YELLOW EDGES First turn U so at least two top edges match the center under them. This algorithm is called Sune; you use it on every card after this one.  two matched, side by side hold them BACK and LEFT RUR'U' RU2R'  two matched, opposite run it once from any hold, then look again RUR'U' RU2R' LAST-LAYER ORDER ON THIS CARD: yellow cross, match edges, place corners, twist corners. Card 2 replaces this order permanently – it is the only thing on this card that gets retired. UNLOCK CARD 2 – five scrambles in a row with this card face down. ◻ MASTER: those five under 3:00. My target: _____. -NEXT: 2/3 TWO-LOOK  80 mm
TWO-LOOK PLL CORNERS -headlights = two matching corners on one face  T headlights on ONE face – hold that face LEFT RUR'U' R'F R2UR'U' RUR'F'  Y no headlights – the two swapping corners sit diagonally FRUR'U' RUR'F' RUR'U' R'FRF' PLL EDGES -corners home, top still not solved  Ub the front edge travels LEFT R2U RUR'U' R'U' R'UR'  Ua the front edge travels RIGHT M2U MU2 M'UM2  H both pairs swap straight across M2UM2 U2 M2UM2  Z each pair swaps with its neighbour M'U' M2UM2U' M'U2 M2U THE NEW ORDER: yellow cross, yellow face, corners home, edges home. This never changes again. Card 1's order is retired, and so is Niklas: it moves corners but wrecks the yellow face, and here the yellow face is finished first. UNLOCK CARD 3 – fifteen last layers in a row, each finished in exactly two algorithms, case named out loud first. A repeat does not count. ◻ MASTER: solves averaging under 1:00. My target: _____. -NEXT: 3/3 ONE-LOOK PLL  80 mm	TWO-LOOK PLL CORNERS -headlights = two matching corners on one face  T headlights on ONE face – hold that face LEFT RUR'U' R'F R2UR'U' RUR'F'  Y no headlights – the two swapping corners sit diagonally FRUR'U' RUR'F' RUR'U' R'FRF' PLL EDGES -corners home, top still not solved  Ub the front edge travels LEFT R2U RUR'U' R'U' R'UR'  Ua the front edge travels RIGHT M2U MU2 M'UM2  H both pairs swap straight across M2UM2 U2 M2UM2  Z each pair swaps with its neighbour M'U' M2UM2U' M'U2 M2U THE NEW ORDER: yellow cross, yellow face, corners home, edges home. This never changes again. Card 1's order is retired, and so is Niklas: it moves corners but wrecks the yellow face, and here the yellow face is finished first. UNLOCK CARD 3 – fifteen last layers in a row, each finished in exactly two algorithms, case named out loud first. A repeat does not count. ◻ MASTER: solves averaging under 1:00. My target: _____. -NEXT: 3/3 ONE-LOOK PLL  80 mm
ONE-LOOK PLL EDGES ONLY -corners already home  Ua B solid, F L R headlights; 3 edges counter-clockwise M2U MU2 M'UM2  Ub B solid, F L R headlights; 3 edges clockwise R2U RUR'U' R'U' R'UR'  Ha all 4 headlights; opposite edges swap M2U'M2 U2 M2U'M2  Za all 4 headlights; adjacent edges swap M'U' M2U'M2U' M'U2 M2U HOW TO LOOK - Count the headlights faces. One = front of this card. Four = this column. None = the column on the right. - Only then read the edges to pick the exact case. - Line the case up as drawn, run it, turn the top again to finish – that last free turn is the AUF. • already yours from Card 2. WORDS PLL = slide the top pieces home - headlights = two matching corners on one face - AUF = the free top turn that lines a case up - adjacent corner swap = the swapping corners share an edge - diagonal corner swap = the swapping corners sit opposite Card 2 gives you 19/72 last layers in one look. This card gives 71/72. Next: the other 57 top-face cases, and the first two layers solved as pairs – drilled with a cube in the app: cubepath-six-vercel-app/practice DONE – all twenty-one named correctly, twenty-one in a row, no card in hand, on two separate days. ◻ MASTER: PLL under 3 s, solves under 0:30. My target: _____. -NEXT: 3/3 ONE-LOOK PLL  80 mm	ONE-LOOK PLL EDGES ONLY -corners already home  Ua B solid, F L R headlights; 3 edges counter-clockwise M2U MU2 M'UM2  Ub B solid, F L R headlights; 3 edges clockwise R2U RUR'U' R'U' R'UR'  Ha all 4 headlights; opposite edges swap M2U'M2 U2 M2U'M2  Za all 4 headlights; adjacent edges swap M'U' M2U'M2U' M'U2 M2U HOW TO LOOK - Count the headlights faces. One = front of this card. Four = this column. None = the column on the right. - Only then read the edges to pick the exact case. - Line the case up as drawn, run it, turn the top again to finish – that last free turn is the AUF. • already yours from Card 2. WORDS PLL = slide the top pieces home - headlights = two matching corners on one face - AUF = the free top turn that lines a case up - adjacent corner swap = the swapping corners share an edge - diagonal corner swap = the swapping corners sit opposite Card 2 gives you 19/72 last layers in one look. This card gives 71/72. Next: the other 57 top-face cases, and the first two layers solved as pairs – drilled with a cube in the app: cubepath-six-vercel-app/practice DONE – all twenty-one named correctly, twenty-one in a row, no card in hand, on two separate days. ◻ MASTER: PLL under 3 s, solves under 0:30. My target: _____. -NEXT: 3/3 ONE-LOOK PLL  80 mm
ANNEX — BIG CUBES AND THE MAP THE LADDER 1/3 FIRST SOLVE - 9 algorithms - unlock: five solves, card face down. 2/3 TWO-LOOK - +13 - unlock: fifteen last layers in exactly two algorithms. 3/3 ONE-LOOK PLL - +15 - done: all 21 named, twice on separate days. After that: full QLL, F2L, cross planning, look-ahead – in the app. WHY THERE ARE ONLY THREE A card is good at a closed set of cases you tell apart by sight. It is bad at a skill you drill. F2L, cross planning and look-ahead are drills with no finite case list. Full QLL is 57 cases, 22 of which differ only in a silver a fifth of a millimetre wide at this size. All of them are in the app, with a real cube and randomised setup. This set stops where the medium stops. PRINT AND VERSION Every ALGORITHM on these cards is expanded from machine-verified data and checked against a cube simulator at build time; if one were wrong the build would fail rather than ship. Recognition wording is generated from the same permutation diagram is drawn from. Set v1.0 - reprint any single card at cubepath-six-vercel-app/print - print at 100% / Actual size, never fit to page. Not a tier, no unlock, no order. Use it when you buy a 4x4.  80 mm	ANNEX — BIG CUBES AND THE MAP THE LADDER 1/3 FIRST SOLVE - 9 algorithms - unlock: five solves, card face down. 2/3 TWO-LOOK - +13 - unlock: fifteen last layers in exactly two algorithms. 3/3 ONE-LOOK PLL - +15 - done: all 21 named, twice on separate days. After that: full QLL, F2L, cross planning, look-ahead – in the app. WHY THERE ARE ONLY THREE A card is good at a closed set of cases you tell apart by sight. It is bad at a skill you drill. F2L, cross planning and look-ahead are drills with no finite case list. Full QLL is 57 cases, 22 of which differ only in a silver a fifth of a millimetre wide at this size. All of them are in the app, with a real cube and randomised setup. This set stops where the medium stops. PRINT AND VERSION Every ALGORITHM on these cards is expanded from machine-verified data and checked against a cube simulator at build time; if one were wrong the build would fail rather than ship. Recognition wording is generated from the same permutation diagram is drawn from. Set v1.0 - reprint any single card at cubepath-six-vercel-app/print - print at 100% / Actual size, never fit to page. Not a tier, no unlock, no order. Use it when you buy a 4x4.  80 mm

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